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Semester:	Winter 2025
Class:	MTH 176
Section:	07
Exam:	1 (Make-up)
Document #:	/ 1

Washtenaw Community College — MTH 176 College Algebra

Midterm Exam 1 (Make-up)

Feb 17 – Feb 19, 2025

Name:

Instructions for the exam

- You will have **90 minutes** to complete the exam.
- There are **7 problems**, some of which may have multiple parts. Each problem is worth a total of **10 points**.
- A graphing calculator is *required* for this exam.
- **This is a closed-book exam.** You may not refer to external sources such as notes (including the course notes) and books, and you are not allowed to receive help from anyone else.

However, you *are* allowed to use a notecard for the exam. For the notecard, you can only use **at most one sheet** with maximum dimensions of **5 inches by 8 inches**. You may use both sides of the notecard to fill as much material you'll reference during the exam.

- **No work, no credit!** Unless otherwise indicated, every problem requires work to be shown; that is, you'll need to give a valid explanation or process along with your answer. Such problems that include correct answers with no valid explanation will not be given any credit. Incorrect or incomplete answers where some correct work is shown are likely to receive some partial credit.
- **Need space to show work?** Each problem page has remaining space for your work and solution; this includes the empty back page of the problem page. If there is work that you do not want to be counted for grading, cross it out with a large "X." If you use an additional page for any problem, please write a note on the problem page (example: "Go to page 4" or "Go to scrap paper") telling the grader where to look, and label your work on those pages for the designated problem as necessary (example: to the left of your work for Problem 4(a), write "4(a)").

You may ask for extra scrap paper as provided by the Testing Center or Learning Support Services, and whatever scrap paper you have used will be collected and attached to the exam. Your work on scrap paper will be not counted for grading, *unless you indicate (on the original problem page) the grader to look over there, in order for your work to be counted*. Thus, even if you have a complete and correct explanation for your solution on scrap paper (and not anywhere else on the exam paper), you will **not** receive any credit if you don't redirect the grader to the respective scrap paper page(s) for that explanation.

Good luck!

Grading

Problem 1	/ 10 points
Problem 2	/ 10 points
Problem 3	/ 10 points
Problem 4	/ 10 points
Problem 5	/ 10 points
Problem 6	/ 10 points
Problem 7	/ 10 points
Total	/ 70 points

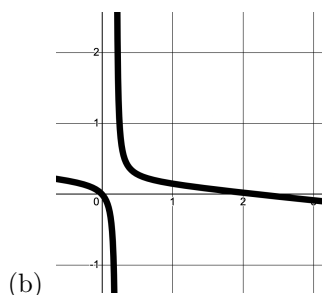
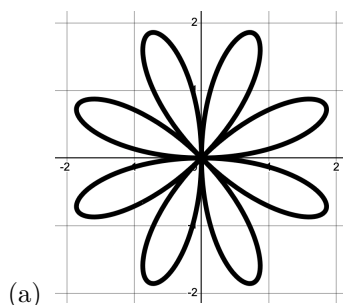
Problem 1 [10 points]

- (a) **(2 points)** Consider two points $P = (\frac{3}{2}, -4)$ and $O = (-\frac{1}{2}, -8)$. Determine $\text{distance}(O, P)$.
- (b) **(3 points)** Write the equation of a circle centered at O that passes through the point P . (Pay attention to what's being identified as the center!)
- (c) **(5 points)** Now consider the equation $x^2 - 5x + y^2 + 4y = -\frac{5}{4}$. This defines a circle. Identify the center and radius of this circle that this equation defines.

Problem 2 [10 points]

Consider the following relations, either defined as a graph or as an equation. For each of these:

- (i) Is this a function? If it is, you do not have to explain why. If it is *not*, explain why.
- (ii) If this is a function, is this one-to-one? *You must explain why if it is one-to-one, or if it is not.* (If this relation was not a function in the first place, you do not need to answer this part.)



(c) $|x + 4| + y = -1$

(d) $x^2 + (y - 2)^2 = 1$

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Problem 3 [10 points]

Consider the following functions

$$f(x) = 7 - 9x^2, \quad g(x) = \begin{cases} -\frac{2}{x+2} + 3 & \text{if } x < -2 \\ 8 - 3|x - 2| & \text{if } -2 \leq x < 3, \\ 5 & \text{if } x > 3 \end{cases}, \quad h(x) = \frac{x\sqrt{5+2x} + |x|}{x^2 - 4}.$$

- (a) **(2 points)** Compute $g(-2)$.
- (b) **(4 points)** Compute $\frac{f(3+h)-f(3)}{h}$. (Leave your answer in terms of h , and try to simplify as much as possible.)
- (c) **(4 points)** Compute the domain of h . Write your answer as an interval or as a union of intervals.

Problem 4 [10 points]

(For this part, you will need to use a graphing calculator, and therefore you do not need to show any work. Round your answers to the nearest hundredth.)

Let $g(x) = 4x^8 + 5x^6 - 11x^5 - 6x^2 + 2$. Figure out the following for g :

- (a) **(1 point)** Its zeros
- (b) **(1 point)** Its y -intercept(s)
- (c) **(2 points)** Its absolute extrema (here, name the *points* where the function attains its absolute maximum or minimum values, and label as such)
- (d) **(2 points)** Its local extrema (here, name the *points* where the function attains its absolute maximum or minimum values, and label as such)
- (e) **(1 point)** The set of inputs where the function is increasing
- (f) **(1 point)** The set of inputs where the function is decreasing
- (g) **(1 point)** Its domain
- (h) **(1 point)** Its range

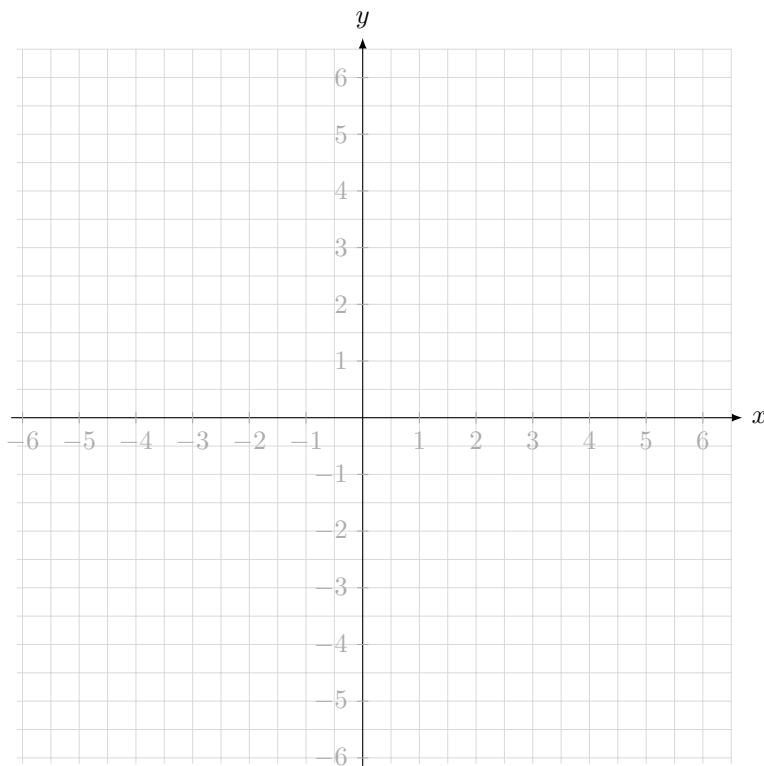
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Problem 5 [10 points]

(a) Consider the function $h(x) = 3 - (x + 1)^2$.

- (i) **(2 points)** Identify the zeros of h . Give both **exact** and **rounded answers to the nearest hundredth** for this part.
- (ii) **(4 points)** Observe that h is the result of a combination of transformations on the squaring function $q(x) = x^2$. Identify the transformations used to transform the squaring function q into the function h (for instance, write down “Dilate horizontally by a factor of $\frac{1}{3}$ ” if that is a relevant transformation involved to make h).

Then, use those transformations to graph $y = h(x)$ in the Cartesian plane below as accurately as possible. (It is okay to check if your sketch is correct using your graphing calculator for this part!)



(b) Let $f(x) = \sqrt{3 - x}$ and $g(x) = (x + 1)^2$.

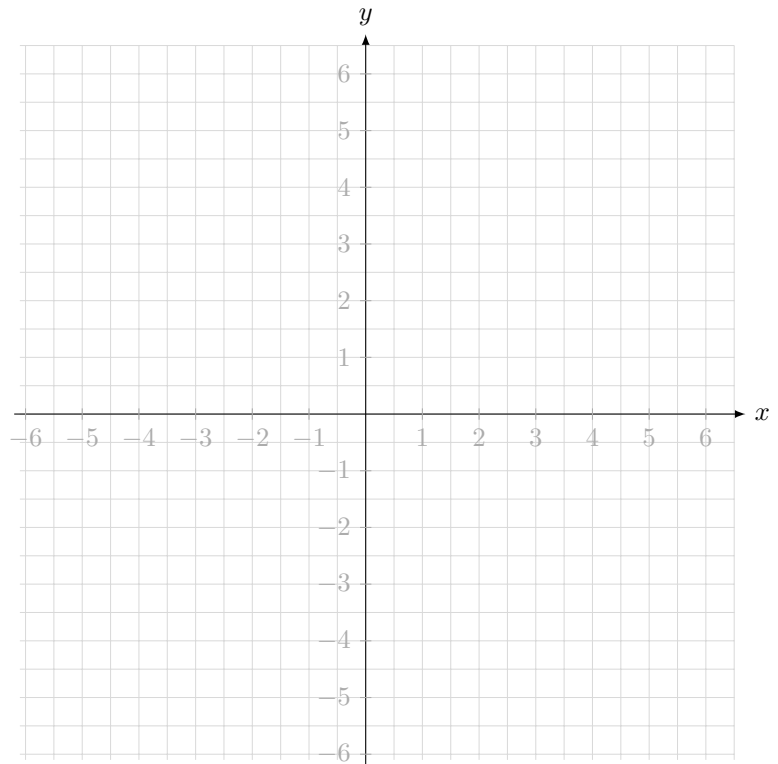
- (i) **(2 points)** What is the domain of $f + g$?
- (ii) **(1 point)** Compute the composition $(f \circ g)(x)$. (You do not need to expand anything, like products of binomials, in your final answer.)
- (iii) **(1 point)** What is the domain of $f \circ g$? (Hint: You'll need your graph of h and its zeros from part (a). For what input values of h is the graph at or above the x -axis? That should tell you about $\text{domain}(f \circ g)$. Be sure to use **exact numbers** for the endpoints of your domain intervals.)

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Problem 6 [10 points]

Let $y = f(x)$ be the function defined by $2y + 3x = 12$.

- (a) **(2 points)** Find the intercepts of the function f .
- (b) **(2 points)** Determine the slope of f .
- (c) **(4 points)** Based on the previous two parts, rewrite $y = f(x)$ in slope-intercept form and graph the function f in the Cartesian plane below. (Though, it's okay if you already rewrote $y = f(x)$ in slope-intercept form to answer any of the previous two parts; just make sure it's correct, as your work for both the intercepts and slope will doubly-count for both parts.)

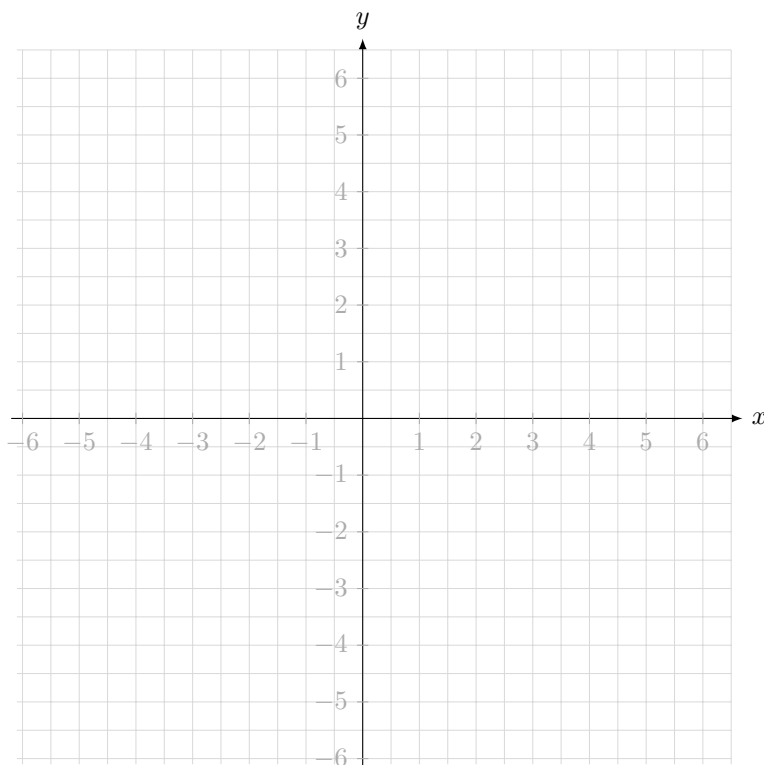


- (d) **(2 points)** Having found out what $f(x)$ is from the previous part, we observe that it is one-to-one. Determine its inverse $f^{-1}(x)$.

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Problem 7 [10 points]

- (a) **(2 points)** Solve the following equation: $|x - \frac{5}{2}| = 3$.
- (b) **(2 points)** Based on your answer from the previous part, what is the solution set to the inequality $|x - \frac{5}{2}| > 3$? (Remember to show how you got that solution set!)
- (c) **(4 points)** Rewrite the following function $f(x) = |x - 1| - |2x + 1|$ as a piecewise-defined function.
- (d) **(2 points)** Sketch the graph of f in the Cartesian plane below, based on the previous part.



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